

Aequitas

Node Operator Guide — How to Join the Network

What is an Aequitas Node?

Every node in the Aequitas network independently validates the BlockDAG, synchronises with peers, and helps keep the Proof-of-Humanity ledger decentralised. Running a node contributes to the security and resilience of the network — and, once the validator-pool distribution is active, node operators receive a share of protocol fees.

Requirements

Component	Minimum	Recommended
OS	Ubuntu 22.04 LTS	Ubuntu 24.04 LTS
RAM	1 GB	2 GB
Disk	10 GB SSD	20 GB SSD
CPU	1 vCPU	2 vCPU
Network	10 Mbit/s	100 Mbit/s
Go version	1.21+	1.22+
Outbound port	4001 (P2P)	4001 + 8080 (API)

Step 1 — Clone the Repository

Clone the official Aequitas repository from GitHub:

```
git clone https://github.com/hanoi96international-gif/Aequitas.git cd
Aequitas/aequitas-chain
```

Step 2 — Build the Node Binary

Compile the Go binary. This creates the *aequitasd* executable:

```
go build ./cmd/aequitasd/
```

If the build fails, make sure you have Go 1.21+ installed: go version

Step 3 — Configure Environment Variables

Create a `.env` file or export these variables in your shell before starting the node:

Variable	Value	Description
DATABASE_URL	postgres://...	PostgreSQL connection string

PEER_NODES	https://aequitas-production-9fba.up.railway.app	Bootstrap peer (official node)
IS_PRIMARY_NODE	false	Set to true ONLY on one designated node for UBI payouts
PORT	8080	HTTP API port (optional, default 8080)

Important: Only ONE node should have IS_PRIMARY_NODE=true — this node runs the daily UBI pool distribution. If multiple nodes have this set, the pool would be distributed multiple times per day.

Step 4 — Start the Node

Run the node directly:

```
./aequitasd
```

Or as a background process:

```
nohup ./aequitasd > aequitas.log 2>&1 & tail -f aequitas.log
```

Step 5 — Verify Your Node is Syncing

After startup, your node will automatically discover and sync from the bootstrap peer. You should see messages like this in the logs:

```
[HTTP-SYNC] Added 50 new blocks from https://aequitas-production-9fba.up.railway.app [DAG]
Added peer block #68260 | Tips: 2 [Block #68261] Hash: 0910c4... | Humans: 1 | Time:
17:55:21
```

Once synced, check your node's API: `curl http://localhost:8080/api/status`

Deploying to a Cloud Provider

Aequitas nodes run well on Render (free tier) or Railway. The official secondary node runs on Render.

Render (render.com)

Setting	Value
Service Type	Web Service
Build Command	go build -o aequitasd ./cmd/aequitasd/
Start Command	./aequitasd
Instance Type	Free (512 MB RAM)
Health Check Path	/api/status

Network Information

Parameter	Value
Chain ID	1926 (0x786)

Block Time	~6 seconds
Consensus	BlockDAG (Directed Acyclic Graph)
V7 Contract	0xE832Ac8Fa64F1AE2c6a5fE5d7DFbF0f9475ec0ae
Bootstrap Node	https://aequitas-production-9fba.up.railway.app
Secondary Node	https://aequitas-node-2.onrender.com
GitHub	github.com/hanoi96international-gif/Aequitas

Aequitas — Money exists because people exist. Nothing more, nothing less.

aequitas-production-9fba.up.railway.app